

Supplementary Material:

An End-to-End Multimodal System for Subtitle Recognition and Chinese–Japanese Translation with Speech Synthesis in Short Dramas

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Algorithm 1 Confidence-Adaptive OCR/ASR Fusion

Require: Whisper segments $\mathcal{A}$, OCR detections $\mathcal{O}$, time tolerance δ , log-prob threshold θ_p , similarity threshold θ_s

Ensure: Fused subtitle sequence $\mathcal{S}$

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1:  $\mathcal{S} \leftarrow \emptyset$ 
2: for each  $a_i = (t_i^s, t_i^e, w_i, p_i) \in \mathcal{A}$  do
3:    $\mathcal{C}_i \leftarrow \{o_j \in \mathcal{O} \mid \tau_j \in [t_i^s - \delta, t_i^e + \delta]\}$   $\triangleright$  candidate OCR detections
4:   if  $p_i \geq \theta_p$  then  $\triangleright$  ASR is confident; trust Whisper
5:      $\hat{w}_i \leftarrow w_i$ 
6:   else
7:      $o^* \leftarrow \arg \max_{o_j \in \mathcal{C}_i} \text{sim}(o_j, w_i)$   $\triangleright$  best OCR match
8:     if  $\mathcal{C}_i \neq \emptyset$  and  $\text{sim}(o^*, w_i) \geq \theta_s$  then
9:        $\hat{w}_i \leftarrow o^*$   $\triangleright$  substitute OCR text, retain ASR timing
10:    else
11:       $\hat{w}_i \leftarrow w_i$   $\triangleright$  fallback: keep ASR
12:    end if
13:  end if
14:   $\mathcal{S} \leftarrow \mathcal{S} \cup \{(t_i^s, t_i^e, \hat{w}_i)\}$ 
15: end for
16: return  $\mathcal{S}$ 

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TABLE I
STATISTICAL INFORMATION OF THE PROPOSED DATASET.

Dataset Composition		Subtitle Statistics	
Video clips (total)	82	Total duration (min)	130.56
Paired episodes	79	Subtitle segments	3,692
Speaker IDs	12	Avg. segments/episode	46.7

This document collects supporting tables, a figure, and the fusion pseudocode referenced from the main manuscript. Numbering is independent of the main paper.

TABLE II

GATING-RULE ABLATION UNDER A SINGLE UNIFIED EVALUATION HARNESS (IDENTICAL FUSION IMPLEMENTATION AND CORPUS-LEVEL SEGMENT-JOINED METRICS; ALL 79 EPISODES). “STATIC” ROWS OPEN THE LOG-PROBABILITY GATE FOR EVERY SEGMENT ($\theta_p=0$); THE ADAPTIVE ROW APPLIES THE SELECTED OPERATING POINT ($\theta_p = -0.2$, $\theta_s=80$). ALL ROWS ARE DIRECTLY COMPARABLE. ABSOLUTE VALUES DIFFER FROM THE PRODUCTION-PIPELINE FUSION METRICS IN THE MAIN PAPER DUE TO THE CORPUS-LEVEL TEXT-JOINING FORMAT. BEST PER COLUMN IN BOLD.

Fusion Strategy	BLEU $\uparrow$	chrF++ $\uparrow$	CER $\downarrow$	CharAcc $\uparrow$	Comp. $\uparrow$
Never Replace (ASR only)	81.31	67.20	0.156	0.898	0.8066
Always Replace ($\theta_s=0$)	0.51	14.07	0.872	0.138	0.1030
Static ($\theta_s=40$)	81.35	68.32	0.161	0.901	0.8091
Static ($\theta_s=60$)	81.48	68.12	0.158	0.900	0.8094
Static ($\theta_s=80$)	81.61	67.77	0.156	0.899	0.8092
Adaptive ($\theta_p = -0.2$, $\theta_s=80$, ours)	81.60	67.73	0.156	0.899	0.8091

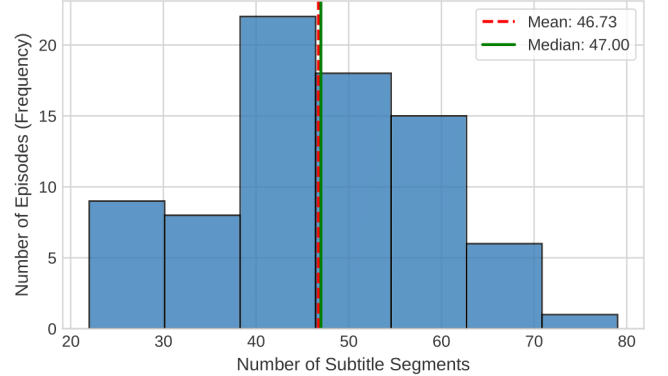


Fig. 1. Distribution of subtitle segments per episode across the 79-episode dataset.

TABLE III
FRAME SAMPLING RATE ABLATION FOR SUBTITLE RECALL ON EPISODE 11.MP4 USING QWEN3-VL-4B.

Sampling Rate	Detected Characters	Recall vs. 5 fps
1 fps	411	27.7%
2 fps	659	44.3%
5 fps	1,486	100% (upper bound)

TABLE IV

GRID SEARCH COMPOSITE SCORES ON THE 20-EPISEODE VALIDATION SET. EACH CELL IS THE CORPUS-LEVEL COMPOSITE SCORE FOR THE FUSION STRATEGY GATED BY (θ_p, θ_s) . THE UPPER-LEFT “FLAT” REGION ($\theta_p \leq -0.8$) CORRESPONDS TO THE ASR-ALWAYS REGIME: ONLY 1.6% OF CORPUS SEGMENTS HAVE AVG LOGPROB BELOW -0.8 , SO ESSENTIALLY NO OCR SUBSTITUTIONS OCCUR. SCORES ARE ROUNDED TO THREE DECIMAL PLACES; THE BOLD MARKS THE UNIQUE FULL-PRECISION MAXIMUM ($\theta_p = -0.2$, $\theta_s = 80\%$: 0.8051 vs. 0.8046 AT $\theta_s \in \{40\%, 70\%\}$, WHICH ALSO ROUND TO 0.805).

$\theta_p \setminus \theta_s$	40%	50%	60%	70%	80%
≤ -0.8	0.800	0.800	0.800	0.800	0.800
-0.5	0.802	0.801	0.800	0.800	0.800
-0.4	0.802	0.801	0.800	0.800	0.800
-0.3	0.804	0.803	0.802	0.802	0.803
-0.2	0.805	0.803	0.803	0.805	0.805

TABLE V

WORD-LEVEL ERROR TYPE ANALYSIS OF ADAPTIVE FUSION CORRECTIONS (JIEBA POS; NR/NS/NT/NZ = PROPER NOUNS). GT CORRECTION RATE: FRACTION OF SUBSTITUTIONS THAT MOVED THE OUTPUT CLOSER TO THE GROUND TRUTH.

Category	Substitutions	% of Total	GT Corrections	Correction Rate
Proper nouns (nr/ns/nt/nz)	30	13.2%	18	60.0%
Numbers	4	1.8%	3	75.0%
Common vocabulary	193	85.0%	114	59.1%
Total	227			

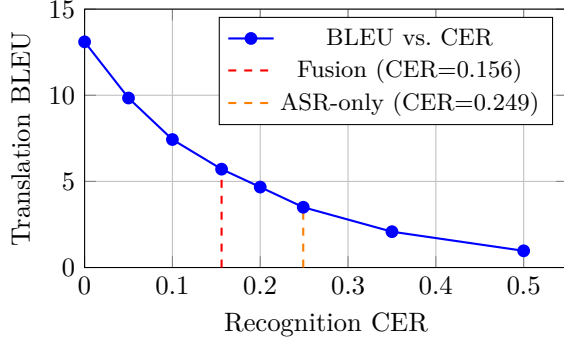


Fig. 2. Cross-stage error propagation: translation BLEU as a function of recognition CER. Dashed lines mark the operating points of ASR-only (CER=0.249) and Adaptive Fusion (CER=0.156). The fusion’s recognition improvement yields +2.2 BLEU in the translation stage.

TABLE VI

END-TO-END PIPELINE EVALUATION ON 16 FOLD-0 VALIDATION EPISODES. SOURCE TEXT FED TO THE QWEN3-VL-4B LoRA TRANSLATION MODEL VARIES BY CONDITION; METRICS ARE COMPUTED AT THE DOCUMENT LEVEL (ENTIRE EPISODE AS ONE UNIT) AGAINST THE JAPANESE GROUND TRUTH. CHRF++ IS THE PRIMARY METRIC (CHARACTER-LEVEL, MORE ROBUST THAN BLEU FOR JAPANESE AT DOCUMENT GRANULARITY). THE ASR-ONLY BLEU OF 0.00 IS A GENUINE CONSEQUENCE OF PROPAGATED PROPER-NOUN ERRORS ELIMINATING ALL DOCUMENT-LEVEL 4-GRAM MATCHES, NOT AN EVALUATION ARTEFACT (SEE TEXT).

Source Condition	BLEU $\uparrow$	chrF++ $\uparrow$
Oracle (gold Chinese text)	5.02	14.97
Adaptive Fusion (ours)	0.48	13.56
ASR-only (Whisper-medium)	0.00	11.32

TABLE VII

TRANSLATION QUALITY ABLATION: VISUAL CONTEXT VS. TEXT-ONLY INPUT USING QWEN3-VL ZERO-SHOT ON 20 EPISODES. ABSOLUTE BLEU GAIN: +3.35.

Input Modality	BLEU $\uparrow$	chrF++ $\uparrow$
Text-Only (OCR subtitles)	10.09	28.26
Multimodal (Video frames + Text)	13.44	29.70

TABLE VIII

TTS REFERENCE AUDIO DURATION ABLATION USING F5-TTS. SPEAKER SIMILARITY MEASURED VIA COSINE SIMILARITY OF SPEAKER EMBEDDINGS.

Reference Duration	Speaker Similarity $\uparrow$
Short (~ 3 s)	0.64
Long (~ 10 s)	0.71

TABLE IX

QUALITATIVE TRANSLATION EXAMPLES. EXAMPLE A SHOWS A CULTURALLY NUANCED TERM TRANSLATED CORRECTLY BY FT BUT INCORRECTLY BY ZS. EXAMPLE B SHOWS AN OOV-DRIVEN HALLUCINATION BY FT THAT ZS AVOIDS.

Ex.	Chinese source	FT (ours)	ZS (Qwen3-VL-4B)	Reference
A	看来我们挺有缘	きっと縁があるな (cultural term preserved)	きっと運が良いのだろう (semantic error: 縁 $\rightarrow$ 運)	やっぱり縁があるのかもね
B	拜托	お前はお母さんも有名なファッションデザイナーの White さんだろ (OOV hallucination)	お祈り	っていうか